Learning Objective 1.3: I can describe signal-space concepts (basis functions, constellations, decision regions) and apply them to a digital modulation scheme.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Constellations and data rate.** A link must carry 1 Mb/s.

   (a) How many bits per symbol do 8-PSK and 64-QAM carry?

   8-PSK: ____

   64-QAM: ____

   (b) Find the symbol rate the link needs with QPSK and with 16-QAM.

   QPSK: ____

   16-QAM: ____

   (c) Sketch QPSK and 4-PAM at the same average symbol energy, and explain in one or two sentences why a radio uses QPSK rather than 4-PAM on a carrier.

2. **Noise and decisions.** A BPSK receiver sees its two points at $+1$ V and -1 V on the I axis. The noise on each axis is Gaussian with a standard deviation of 0.4 V. Use $Q(2.5) = 6.2 \times 10^{-3}$ and $Q(4) = 3.2 \times 10^{-5}$.

   (a) State the decision rule and draw the decision regions.

   (b) Find the bit error rate.

   P_b =

   (c) The noise drops to 0.25 V. Find the new bit error rate, and the change in SNR in dB.

   P_b =

   ΔSNR =

   (d) A 4.1 dB improvement in SNR cut the error rate by a factor of about 200. Explain why the bit error rate changes so much faster than the SNR.

3. **Eb/N0, Es/N0, and SNR.** A receiver filtered to its symbol rate measures an SNR of 15 dB.

   (a) Find E_s/N_0, and E_b/N_0 for QPSK and for 16-QAM.

   QPSK:

   16-QAM:

   (b) Use the lesson's BER curves to estimate the bit error rate of each.

   QPSK:

   16-QAM:

   (c) Explain why BER curves are plotted against E_b/N_0 rather than SNR.

4. **Gray coding.** An 8-PSK modem places its points every 45° around a circle.

   (a) Label the eight points with a Gray code, starting from 000 at 0°.

   (b) With natural binary labels instead (000, 001, 010, ..., 111), how many bits does a slip from the point labeled 011 to its neighbor 100 corrupt?

   (c) The Gray-coded modem measures a symbol error rate of 3×10^{-4}. Estimate its bit error rate.

   P_b =

5. **Choosing a modulation.** A link budget predicts an SNR of 21 dB in a receiver filtered to 1 Msym/s. The link needs a bit error rate of 10^{-5} and at least 6 dB of margin. At that error rate QPSK needs $E_s/N_0 = 12.6$ dB, 8-PSK needs 17.7 dB, and 16-QAM needs 19.5 dB.

   (a) Find the margin each modulation leaves.

   QPSK: ____

   8-PSK: ____

   16-QAM: ____

   (b) Which modulation meets the requirement, and what data rate does it carry?

   ____

   (c) A customer wants 4 Mb/s. One option is 16-QAM at 1 Msym/s; the other is QPSK at 2 Msym/s, which needs twice the bandwidth at the same transmit power. Compare the margin of each and recommend one.

6. **Reading a constellation.** Four QPSK receivers each show one problem on a constellation sink. Name the cause of each.

   (a) The four points sit in the middle of their quadrants, but each is a round cloud about a third of the way to the boundaries.

   (b) Four tight clouds, each rotated 30° counterclockwise from where it belongs, standing still.

   (c) The points are smeared into a ring of constant radius.

   (d) Explain why the constellation in part (b) would produce a very high bit error rate if the rotation reached 45°, even with no noise.

Documentation: _____